

Adrenalectomy for Secondary Malignancy: Patients, Outcomes, and Indications

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Objective: The goal of this study was to examine a multi-institutional experience with adrenal metastases to describe survival outcomes and identify subpopulations who benefit from adrenal metastasectomy.

Background: Adrenalectomy for metastatic disease is well-described, although indications and outcomes are incompletely defined.

Methods: A retrospective cohort study was performed of patients undergoing adrenalectomy for secondary malignancy (2002–2015) at 6 institutions. The primary outcomes were disease free survival (DFS) and overall survival (OS). Analysis methods included Kaplan-Meier and Cox proportional hazards.

Results: Of 269 patients, mean age was 60.1 years; 50% were male. The most common primary malignancies were lung (n = 125, 47%), renal cell (n = 38, 14%), melanoma (n = 33, 12%), sarcoma (n = 18, 7%), and colorectal (n = 12, 5%). The median time to detection of adrenal metastasis after initial diagnosis of the primary tumor was 17 months (interquartile range: 6–41). Post-adrenalectomy, the median DFS was 18 months (1-year DFS: 54%, 5-year DFS: 31%). On multivariable analysis, lung primary was associated with longer DFS [hazard ratio (HR): 0.49, $P = 0.008$]. Extra-adrenal oligometastatic disease at initial presentation (HR: 1.84, $P = 0.016$), larger tumor size (HR: 1.07, $P = 0.013$), chemotherapy as treatment of the primary tumor (HR: 2.07 $P = 0.027$) and adjuvant chemotherapy (HR: 1.95, $P = 0.009$) were associated with shorter DFS. Median OS was 53 months (1-year OS: 83%, 5-year OS: 43%). On multivariable analysis, extra-adrenal oligometastatic disease at adrenalectomy (HR: 1.74, $P = 0.031$), and incomplete resection of adrenal metastasis (R1 margins; HR: 1.62, $P = 0.034$; R2 margins; HR: 5.45, $P = 0.002$) were associated with shorter OS.

Conclusions: Durable survival is observed in patients undergoing adrenal metastasectomy and should be considered for subjects with isolated adrenal metastases.

Keywords: adrenal metastasectomy, adrenalectomy, cancer, lung cancer, metastasis, metastatic carcinoma, oligometastases

(*Ann Surg* 2021;274:1073–1080)

The role of surgery in the setting of metastatic cancer has historically been limited to diagnostic and palliative procedures. Advances in systemic chemotherapy and immunotherapy, and improvements in surgical and perioperative care have extended the surgical indications for metastatic disease. Essential to modern understandings of cancer progression is the concept of oligometastatic disease, in which metastases may be limited to individual sites and potentially amenable to curative treatment.¹ A large experience with metastasectomy in specific clinical settings including hepatic metastases for colorectal cancer^{2–5} has encouraged the application of surgery in the oligometastatic setting for other cancers.⁶

The adrenal glands are frequent sites of metastatic disease. Autopsy series demonstrate metastases to the adrenals in 3% of subjects; in patients with known malignancies, adrenal lesions are metastatic in approximately 75%.^{7,8} Adrenal metastasectomy has been well-described in the literature, although survival benefit has not been definitively established in most primary tumor types, and indications are incompletely defined. Early studies focused on resection of isolated metastases. The data is most convincing in non-small cell lung cancer, where resection is superior to medical management of isolated adrenal metastases.^{7,9} As the prevalence of metastasectomy has expanded, so too have the indications. Increasingly, adrenal metastasectomy is utilized in conjunction with multimodality oncologic therapy for treatment of oligometastatic, extra-adrenal disease. Both laparoscopic and open adrenalectomy approaches are safe and well-tolerated.^{10,11} One recent single-institution study suggests that extra-adrenal metastases do not negatively impact survival in patients who undergo adrenal metastasectomy.¹² These data support the overall trend toward expanding indications for adrenalectomy. Yet both guidelines and studies lag behind current clinical practice.

Despite the increasing prevalence of adrenal metastasectomy, data to support this practice are limited. Numerous studies have described single-institution experiences and to identify covariates correlated with survival^{12–21}; few studies have addressed disease-free survival.^{11,12,21,22} No evidence-based guidelines exist for the use of adrenal metastasectomy. The goal of this study was; therefore, 2-fold: first, to identify subpopulations who benefit from adrenal metastasectomy in a multi-institutional cohort; and second, to establish potential expansion of indications for adrenalectomy for metastatic disease.

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A portion of this work was presented in abstract form at the American Association of Endocrine Surgeons annual meeting in Orlando, FL.

This work is a collaborative multi-institutional effort and would not have been possible without the contributions of multiple individuals at each study site. All authors contributed to the study design, data acquisition, manuscript drafting and revision, and all authors approved the final version of the manuscript. Drs. Heather Wachtel and Carrie Lubitz accept direct responsibility for the manuscript.

The authors report no conflicts of interest.

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ISSN: 0003-4932/20/27406-1073

DOI: 10.1097/SLA.0000000000003876

METHODS

Study Population

All patients who underwent adrenalectomy for metastatic cancer at 6 institutions (Brigham and Women's Hospital, Indiana University, Massachusetts General Hospital, Medical College of Wisconsin, University of Pennsylvania, Vanderbilt University) between 2002 and 2015 were identified for inclusion. Retrospective chart review was performed. Demographic, treatment, biochemical, imaging, and pathology data was abstracted from clinical charts. Patients with primary adrenal malignancies, direct adrenal involvement from local tumor invasion, and patients undergoing concomitant ipsilateral nephrectomy for malignancy were excluded. Institutional Review Board (IRB) approval was obtained independently for all study sites.

Definitions

Time to detection of adrenal metastasis was defined as time from diagnosis of the primary tumor to diagnosis of the adrenal metastasis. Consistent with clinical convention, synchronous disease was defined as disease-free interval ≤ 6 months from diagnosis of primary tumor to detection of adrenal metastasis, and metachronous disease as >6 months disease-free interval from primary to adrenal metastasis.^{13,22} Oligometastatic disease was defined as 1–4 sites of disease, consistent with conventional definitions.^{1,23} Adrenalectomy margin status was assessed as R0, R1, or R2 by published criteria.²⁴ Overall survival (OS) was defined as time from adrenalectomy to death. Disease-free survival (DFS) was defined as time from adrenalectomy to disease recurrence, as determined by clinical assessment. Patients who were not rendered disease-free by adrenalectomy (i.e. known additional metastatic disease, or unresected tumor) were considered to have a DFS of 0.

For clarity of treatment time course, therapies were defined with respect to adrenalectomy as the treatment of interest. Chemo-, immuno-, and radiation therapy occurring before adrenalectomy were defined as therapy of the primary tumor, whereas chemo-, immuno-, and radiation therapy occurring after adrenalectomy were defined as adjuvant therapy.

Statistical Analysis

Demographic and clinical data analysis was performed with STATA 15 (Stata Corporation, College Station, TX). Categorical variables were reported as frequency and percentage; continuous variables were reported as mean $\pm$ standard deviation or median with interquartile range (IQR) for normally and non-normally distributed variables, respectively. Group comparisons utilized Student *T*-test for parametric variables, and the Wilcoxon rank-sum (Mann-Whitney *U*) test or Kruskal-Wallis test for nonparametric variables. Kaplan-Meier survival analysis was performed for DFS and OS. Univariate Cox proportional hazards regression analysis was performed to evaluate clinical characteristics associated with DFS and OS. For continuous variables previously described in the literature as associated with survival outcome, Cox regression analysis was performed in 2 independent analyses. First, regression analysis was performed to determine associations between the covariate as a continuous variable and outcomes of DFS and OS. Second, to test previously published correlations, receiver operator characteristic curves were generated, and Area Under the Curve (AUC) values were used to assess the discriminatory value and to assess optimal cut-points for dichotomization. These variables were then tested as binary variables at published cut-points and at independently identified cut-points determined to maximize sensitivity and specificity.^{14,16,18,19,21}

Covariates meeting nominal significance ($P \geq 0.20$) on univariate Cox proportional hazards analysis and covariates of clinical significance

were incorporated into stepwise reduced multivariable models to evaluate factors associated with DFS and OS in independent models. Harrell C-statistic was utilized to assess model goodness-of-fit. The number of covariates included in the model was limited to ensure model stability. A *P*-value of 0.05 was considered statistically significant.

To determine if primary tumor type was associated with survival outcomes, subgroup analyses were performed based upon on primary malignancy type. Tumors were classified as lung primary or non-lung primary, as the most common primary tumor type, and the most extensively studied. Alternatively, tumors were classified as the 5 most common malignancy types (lung, renal cell, melanoma, sarcoma, and colorectal) or other primary tumor.

RESULTS

Study Population Characteristics

Of 269 patients who underwent adrenalectomy for secondary malignancies (Table 1), the mean age was 60.1 years (standard deviation ± 11.1 years) and half were male ($n = 134$, 50%). The most common primary malignancy was lung cancer ($n = 125$, 47%), followed by renal cell carcinoma ($n = 38$, 14%), melanoma ($n = 33$, 12%), sarcoma ($n = 18$, 7%), and colorectal carcinoma ($n = 12$, 5%). Of the 18 sarcoma cases, 14 (78%) were performed at 2 centers. The primary tumor was excised in 74% ($n = 199$) of subjects. Within this group, treatment of the primary tumor consisted of surgery alone ($n = 139$), surgery and chemotherapy ($n = 4$), surgery and radiation ($n = 17$), or surgery, radiation and chemotherapy ($n = 12$). Twenty-seven subjects had incomplete data or missing treatment data. The primary tumor was not surgically excised in 25% of

TABLE 1. Preoperative Characteristics of 269 Patients With Secondary Adrenal Malignancies

Characteristic	N, Median or Mean
Mean age (SD), yr	60.1 (± 11.1)
Male sex	134 (49.8%)
Primary tumor	
Lung	125 (46.5%)
Renal cell	38 (14.1%)
Melanoma	33 (12.3%)
Sarcoma	18 (6.7%)
Colorectal	12 (4.5%)
Esophageal	8 (3.0%)
Breast	6 (2.2%)
Thyroid	5 (1.9%)
Ovarian	3 (1.1%)
Oropharyngeal	3 (1.1%)
Pancreatic neuroendocrine	3 (1.1%)
Endometrial	2 (0.7%)
Cervical	2 (0.7%)
Other primary*	8 (3.0%)
Unknown primary	3 (1.1%)
Median time from primary to metastasis (IQR), months	17 (6–41)
Timing of metastasis with respect to primary tumor	
Synchronous (≤ 6 mo)	70 (26.0%)
Metachronous (>6 mo)	192 (71.4%)
Unknown	7 (2.6%)
Oligometastatic at diagnosis of primary tumor	90 (33.5%)
Oligometastatic at adrenalectomy	80 (29.7%)
Prior metastectomy	55 (20.9%)

*Other primary – tumor types with a single case represented included choriocarcinoma, gastric, hematopoietic, hepatocellular, Merkel cell, prostate, small bowel, and testicular cancers.

IQR indicates interquartile range; SD, standard deviation.

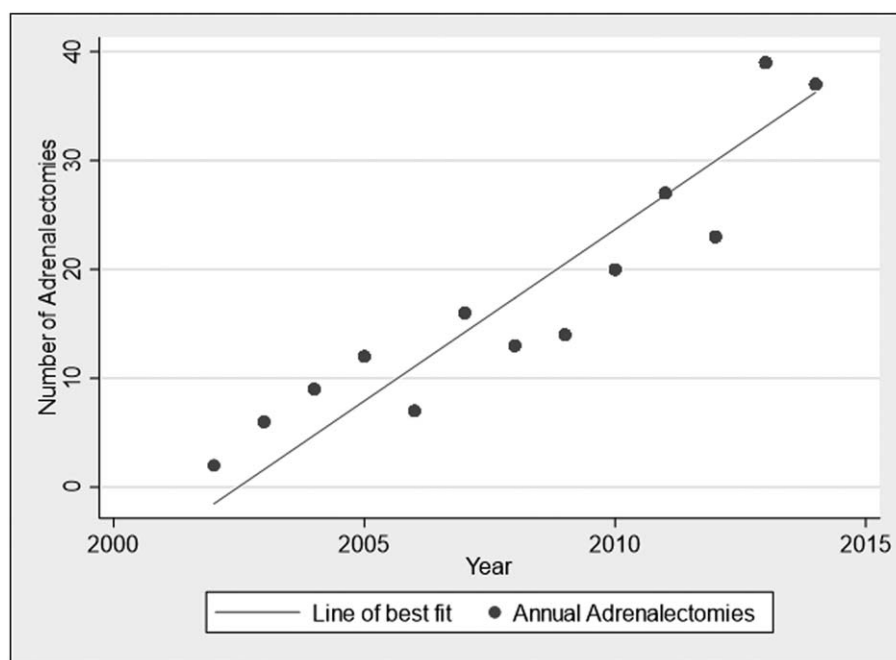


FIGURE 1. Number of adrenalectomies performed annually for metastatic disease, 2002–2015. Line represents linear regression modeling of line of best fit.

patients ($n = 66$). Within this group, treatment of the primary was chemotherapy alone ($n = 9$), immunotherapy alone ($n = 1$), radiation alone ($n = 1$), radiation and chemotherapy ($n = 33$), radiation and immunotherapy ($n = 1$), radiation, chemotherapy, and immunotherapy ($n = 1$), or other therapies ($n = 12$). Eight subjects had incomplete or missing treatment data. The majority (68%) of subjects who did not undergo resection of the primary tumor had lung primaries. From 2002 to 2015, the number of annual adrenalectomies performed for metastatic disease increased (Fig. 1).

Sites of Metastatic Disease

At the time of primary tumor detection, 34% ($n = 90$) of subjects had evidence of oligometastatic disease. Of these, 49 had extra-adrenal metastases alone. Forty-one subjects had adrenal metastases; these were classified as synchronous adrenal metastases. The median time from detection of the primary tumor to detection of the adrenal metastasis was 17 months (IQR: 6–41). The majority of patients were categorized as metachronous disease ($n = 192$, 71%). At the time of adrenalectomy, 21% ($n = 55$) of subjects had undergone prior metastasectomy. Thirty percent ($n = 80$) of patients had evidence of extra-adrenal oligometastatic disease at the time of adrenalectomy. Sixteen of these subjects had previously undergone treatment for extra-adrenal metastases with no residual evidence of disease. Complete data on the indications for adrenalectomy for patients with extra-adrenal disease was not available. However, the most commonly cited reasons for adrenalectomy in the setting of extra-adrenal disease included: planned concomitant or subsequent treatment of the extra-adrenal disease (either with surgical metastasectomy, radiation, or targeted therapies); growth of the adrenal lesion in the setting of otherwise stable extra-adrenal disease; and need to obtain tissue for molecular testing for targeted therapy. Only a very small minority of procedures were performed for purely palliative reasons.

Adrenalectomy and Adjuvant Therapy

Adrenalectomy was performed via laparoscopic transabdominal approach in 51% ($n = 136$), with planned open adrenalectomy in

35% ($n = 93$) and laparoscopic converted to open operation in 8% ($n = 21$); a minority (5%, $n = 13$) of procedures were performed via the posterior retroperitoneoscopic approach. R0 resection margin was achieved in 71% ($n = 190$) with R1 microscopically positive margins in 21% ($n = 57$); only 3% ($n = 7$) had grossly positive R2 margins. The median tumor size was 4.5 cm (IQR: 3–6). Median tumor size was largest in the open surgical group (5.5 cm, IQR: 3.0–8.9), followed by the laparoscopic converted to open group (4.6 cm, IQR: 3.8–5.9), then the laparoscopic transabdominal (4.0 cm, IQR: 2.6–5.6), with the smallest tumors resected by posterior retroperitoneoscopic approach (3.5 cm, IQR: 2.5–4.5), $P = 0.002$. This significant difference suggests that the size of the adrenal metastasis factored into the choice of operative approach, although there was overlap between groups.

Of 252 subjects with complete pathologic margin status, there was no difference in margin status by operative approach ($P = 0.330$). Margins were R0 in 74% of laparoscopic transabdominal, 75% of laparoscopic transabdominal converted to open, 75% of open, and 85% of retroperitoneoscopic surgeries. Margins were R1 in 25% of laparoscopic transabdominal, 25% of laparoscopic transabdominal converted to open, 21% of open, and 8% of retroperitoneoscopic procedures. R2 margins were noted in only 1% of laparoscopic transabdominal, 4% of open, and 8% of retroperitoneoscopic cases. No patients undergoing conversion to open surgery from laparoscopic transabdominal were observed to have R2 margins at final pathology.

Adjuvant chemotherapy was administered after adrenal metastasectomy in 41% ($n = 109$), adjuvant radiation therapy in 20% ($n = 54$), immunotherapy in 13% ($n = 34$), and combined chemotherapy and immunotherapy in 0.7% ($n = 2$). Fifty-eight percent ($n = 157$) had documented disease recurrence. Recurrences were in distant disease sites in 131 patients, locally in the adrenal bed in 8 subjects, and both distant and local recurrence in 18 patients.

Disease-free Survival

The median follow-up time after adrenal metastasectomy in the cohort was 20 months (IQR: 6–41). The median DFS was

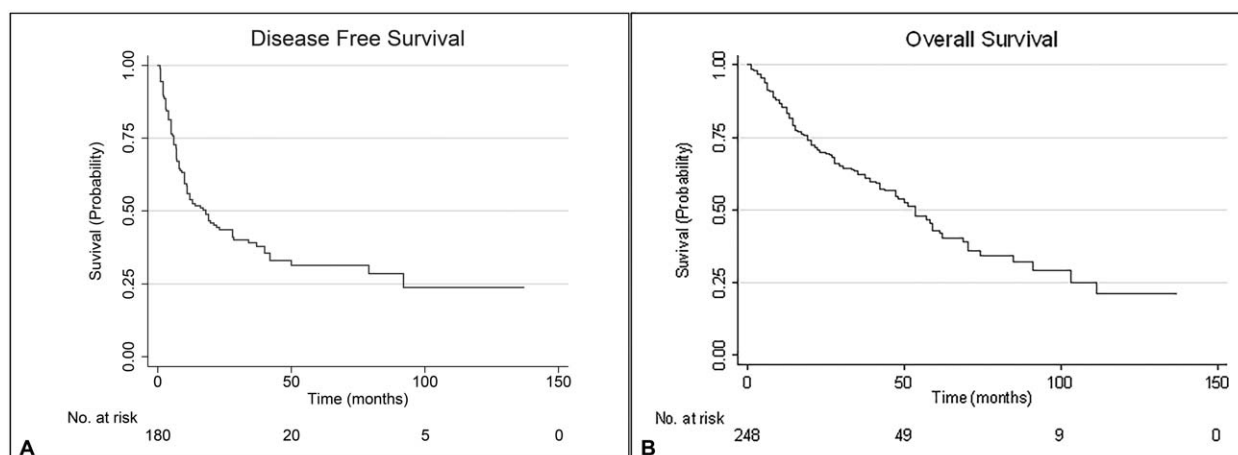


FIGURE 2. Kaplan-Meier survival curves in patients with secondary adrenal malignancies. A, Disease free survival (DFS). Median DFS was 18 months (IQR: 11–28); 1-yr DFS was 54% (95% CI: 0.458–0.612) and 5-yr DFS was 31% (95% CI: 0.231–0.400). B, Overall survival (OS). Median OS was 53 months (IQR: 42–62); 1-yr OS was 83% (95% CI: 0.774–0.875) and 5-yr OS was 43% (95% CI: 0.341–0.516). CI indicates confidence interval; IQR, interquartile range.

18 months (IQR: 11–28); 1-year DFS was 54% [95% confidence interval (CI): 0.458–0.612] and 5-year DFS was 31% (95% CI: 0.231–0.400) (Fig. 2A).

A multivariable Cox regression model was developed to identify covariates associated with DFS (Table 2). Time from primary tumor to adrenal metastasis was examined both as a continuous variable and as a dichotomous variable (synchronous vs metachronous). In the final multivariable model, lung primary

tumor was significantly associated with longer DFS [hazard ratio (HR) 0.49, $P = 0.008$]; however, resection of the primary tumor was not significant. Larger adrenal tumor size (HR 1.07, $P = 0.013$), the presence of extra-adrenal oligometastatic disease at initial presentation (HR 1.84, $P = 0.016$), chemotherapy as definitive treatment of the primary tumor (HR 2.07, $P = 0.027$) and adjuvant chemotherapy (HR 1.10, $P = 0.009$) were significantly associated with shorter DFS.

TABLE 2. Cox Proportional Hazards Regression Analysis Assessing the Association Between Patient Characteristics and Disease Free Survival

	Univariable			Multivariable		
	HR	95% CI	P-value	HR	95% CI	P-value
Age	1.00	0.982–1.019	0.986	—	—	—
Male sex	0.95	0.643–1.400	0.790	—	—	—
Lung primary	0.54	0.357–0.811	0.003	0.49	0.292–0.828	0.008
Primary resection	0.56	0.349–0.903	0.017	—	—	—
Oligometastatic at presentation	1.49	0.988–2.256	0.057	1.84	1.121–3.041	0.016
Prior metastasectomy	0.85	0.519–1.406	0.536	—	—	—
Synchronous	1.05	0.667–1.667	0.820	—	—	—
DFI from primary to metastasis	1.00	0.996–1.004	0.905	—	—	—
Operative approach						
Laparoscopic	Ref	—	—	—	—	—
Laparoscopic converted to open	1.28	0.644–2.541	0.481	—	—	—
Open	1.19	0.776–1.823	0.426	—	—	—
Retroperitoneoscopic	1.23	0.488–3.104	0.660	—	—	—
Size of adrenal tumor	1.05	1.003–1.101	0.036	1.07	1.014–1.124	0.013
Adrenal tumor ≤ 4.5 cm	0.815	0.544–1.220	0.321	—	—	—
Definitive systemic therapy of primary						
Chemotherapy	1.52	0.948–2.444	0.082	2.07	1.086–3.950	0.027
Immunotherapy	7.14	1.686–30.199	0.008	3.07	0.630–14.949	0.165
Chemotherapy + Immunotherapy	—	—	—	—	—	—
Definitive radiation therapy of primary	1.42	0.899–2.248	0.132	1.46	0.798–2.685	0.218
Adjuvant systemic therapy						
Chemotherapy	1.10	0.691–1.736	0.698	1.95	1.185–3.210	0.009
Immunotherapy	1.70	0.968–2.983	0.065	1.89	0.982–3.641	0.057
Chemotherapy + Immunotherapy	2.59	0.351–19.154	0.349	2.132	0.278–16.377	0.467
Adjuvant radiation therapy	1.12	0.702–1.793	0.631	—	—	—

95% CI indicates 95% confidence interval; DFI, disease free interval; HR, hazard ratio.

TABLE 3. Cox Proportional Hazards Regression Analysis Assessing the Association Between Patient Characteristics and Overall Survival

	Univariable			Multivariable		
	HR	95% CI	P-value	HR	95% CI	P-value
Age	1.00	0.978–1.017	0.790	1.00	0.977–1.023	0.991
Male sex	1.15	0.777–1.697	0.489	—	—	—
Lung primary	1.35	0.914–2.000	0.132	1.36	0.831–2.209	0.223
Primary resection	0.52	0.345–0.798	0.003	0.87	0.505–1.509	0.626
Oligometastatic at presentation	1.75	1.178–2.599	0.006	1.37	0.829–2.255	0.220
Oligometastatic at adrenalectomy	2.02	1.328–3.060	0.001	1.74	1.051–2.867	0.031
Prior metastasectomy	0.81	0.500–1.321	0.404	—	—	—
Synchronous	1.51	1.000–2.278	0.050	1.46	0.899–2.379	0.126
DFI from primary to metastasis	0.99	0.985–0.997	0.005	—	—	—
Operative approach						
Laparoscopic	Ref	—	—	—	—	—
Laparoscopic converted to open	0.99	0.501–1.943	0.969	—	—	—
Open	0.95	0.621–1.463	0.827	—	—	—
Retroperitoneoscopic	0.84	0.263–2.705	0.775	—	—	—
Adrenal resection						
R0	Ref	—	—	—	—	—
R1	1.63	1.076–2.458	0.021	1.62	1.038–2.547	0.034
R2	3.26	1.180–9.005	0.023	5.45	1.038–2.547	0.002
Size of adrenal tumor	1.04	0.988–1.098	0.131	—	—	—
Adrenal tumor ≤4.5 cm	0.83	0.558–1.235	0.358	—	—	—
Definitive systemic therapy of primary						
Chemotherapy	1.91	1.242–2.933	0.003	—	—	—
Immunotherapy	1.77	0.244–12.942	0.570	—	—	—
Chemotherapy + Immunotherapy	—	—	—	—	—	—
Definitive radiation therapy of primary	1.82	1.194–2.782	0.005	1.50	0.883–2.559	0.133
Adjuvant systemic therapy						
Chemotherapy	0.78	0.499–1.235	0.295	—	—	—
Immunotherapy	0.60	0.313–1.143	0.120	—	—	—
Chemotherapy + Immunotherapy	0.53	0.073–3.934	0.539	—	—	—
Adjuvant radiation therapy	1.13	0.713–1.801	0.596	—	—	—

95% CI indicates 95% confidence interval; DFI, disease free interval; HR, hazard ratio.

OS

The median OS was 53 months (IQR: 42–62) in the overall cohort. The OS at 1-year was 83% (95% CI: 0.774–0.875) and OS at 5-years was 43% (95% CI: 0.341–0.516) (Fig. 2B). On final age-adjusted Cox-regression multivariable model, oligometastatic extra-adrenal disease at the time of adrenalectomy (HR 1.74, $P = 0.031$), and positive microscopic margins (R1; HR 1.62, $P = 0.034$) or macroscopic margins (R2; HR 5.45, $P = 0.002$) were significantly associated with shorter OS (Table 3). Lung primary tumor was not significantly associated with OS ($P = 0.223$). Resection of the primary tumor ($P = 0.626$), synchronous metastases ($P = 0.126$), and oligometastatic disease at initial presentation ($P = 0.220$) were not significantly associated with OS in the final model.

Survival Outcomes by Primary Tumor Type

To characterize survival outcomes for individual malignancies, we performed subset analysis for the 5 most common primary tumor types (lung, renal cell, melanoma, sarcoma, and colorectal) (Table 4). Shortest median DFS was observed in colorectal carcinoma (7 months), with longest DFS in lung cancer (40 months). Lung cancer; however, was associated with the shortest OS (median 47 months, 5-year survival 36%), followed by renal cell carcinoma (median 59 months, 5-year survival 40%). Greater than two-thirds of patients were alive 5 years after metastasectomy for melanoma (5-year OS 67%), sarcoma (5-year OS 70%), and colorectal cancer (5-year OS 74%). Kaplan-Meier survival curves for DFS and OS are shown in Fig. 3A and B, respectively; patients not included in

TABLE 4. Survival Outcomes of 5 Most Common Primary Tumors

	Disease Free Survival			Overall Survival		
	Median DFS (95% CI)	1-yr DFS (95% CI)	5-yr DFS (95% CI)	Median OS (95% CI)	1-yr OS (95% CI)	5-yr OS (95% CI)
Lung	40 mo (17–NR)	64.4% (0.522–0.743)	43.5% (0.293–0.569)	47 mo (30–57)	79.8% (0.706–0.863)	36.1% (0.245–0.480)
Renal cell	12 mo (7–28)	49.5% (0.282–0.676)	17.0% (0.034–0.395)	59 mo (41–NR)	87.1% (0.690–0.950)	40.3% (0.146–0.652)
Melanoma	12 mo (5–92)	48.7% (0.276–0.669)	33.2% (0.145–0.533)	103 mo (51–NR)	89.8% (0.716–0.966)	67.2% (0.404–0.840)
Sarcoma	37 mo (5–NR)	72.7% (0.371–0.903)	34.6% (0.067–0.661)	NR mo (14–NR)	86.2% (0.550–0.964)	70.4% (0.389–0.879)
Colorectal	7 mo (2–NR)	47.6% (0.123–0.769)	*	61 mo (3–NR)	88.9% (0.433–0.984)	74.1% (0.289–0.930)

*Only 1 patient evaluable at 5-yr time point.

95% CI indicates 95% confidence interval; DFS, disease free survival; NR, not reached; OS, overall survival.

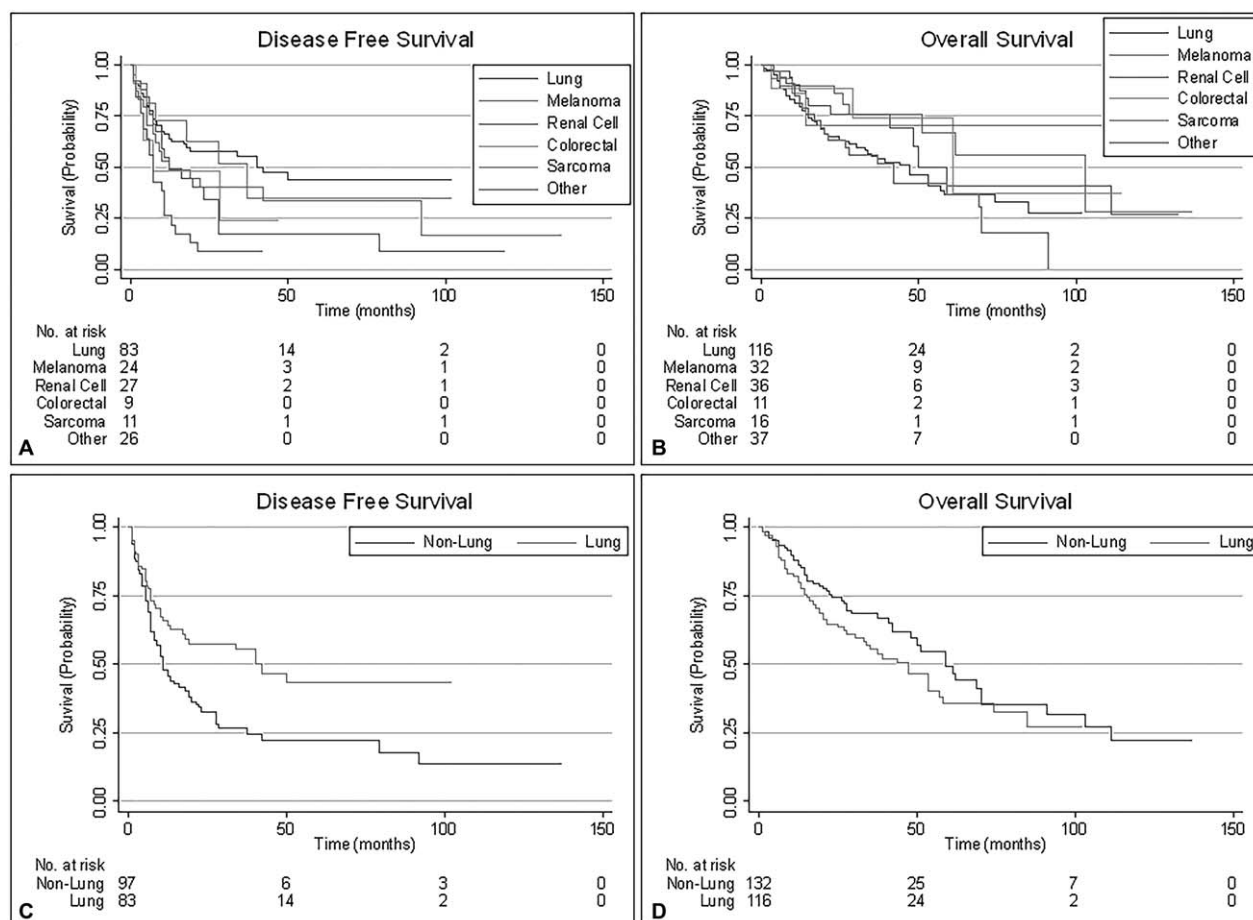


FIGURE 3. Kaplan-Meier survival curves, by tumor type. A, Disease free survival (DFS) is significantly different in the 5 most common primary tumor types ($P < 0.001$); shortest DFS was observed in lung primary tumors. B, Overall survival (OS) is not significantly different in the 5 most common primary tumor types ($P = 0.324$). C, Lung primary tumor is significantly associated with shorter DFS ($P = 0.003$) compared with non-lung primary. D, Lung primary tumor is not significantly associated with shorter OS when compared with non-lung primary tumors ($P = 0.132$).

the 5 most common malignancies were grouped together for comparison.

Lung Primary Tumor

Subjects were separately analyzed by primary malignancy as lung and non-lung primary. There was no significant difference in median duration of clinical follow-up between lung (17 months) and non-lung primaries (23 months, $P = 0.365$). Median DFS was significantly longer in lung (40 months) versus non-lung primary tumors (11 months, $P = 0.003$; Fig. 3C). One and 5-year DFS were 64.4% and 43.5% in lung, and 45.5% and 22.6% in non-lung primaries, respectively. There was no significant difference in median OS (47 vs 59 months, $P = 0.132$) between lung and non-lung primary tumors (Fig. 3D). One and 5-year OS were 79.8% and 36.2% in lung, and 86.2% and 49.6% in non-lung primaries, respectively.

DISCUSSION

In this first multi-institutional series of adrenalectomies performed for secondary malignancies, our data demonstrates that the practice of adrenal metastasectomy encompasses diverse primary tumor types and has increased over time. The expanding utilization of adrenal metastasectomy makes it of paramount importance

to develop data-driven guidelines for the use of this therapeutic strategy.

Our study suggests that durable survival is observed in subjects who undergo adrenal metastasectomy, although more than two-thirds of patients experienced disease recurrence within 5 years of surgery. Disease trajectory seems to be an important predictor of outcomes. We did not observe shorter survival in subjects who initially presented with oligometastatic disease, or subjects who underwent prior metastasectomy. However, the presence of extra-adrenal oligometastatic disease at the time of adrenalectomy was associated with shorter OS. One interpretation of these data is that there is an underlying difference in tumor biology, and successful treatment of independent metastases may be reflective of more indolent disease. Alternatively, this may be a reflection of selection bias, as subjects with prior successfully treated metastases may be referred for adrenalectomy earlier in their course. Unfortunately, complete data on the indications for adrenalectomy for patients with extra-adrenal disease was not available, and therefore in our patient cohort we are unable to comment on this in a statistically meaningful fashion. Nevertheless, there does seem to be a growing trend toward adrenal metastasectomy in the setting of otherwise stable extra-adrenal metastatic disease, as has been noted previously.¹²

It is important to appreciate the distinction between previously treated extra-adrenal metastases, and extra-adrenal metastases present at the time of adrenalectomy. To our knowledge, this is the first study to effectively discriminate between historical and current extra-adrenal metastases. A recent study by Russo et al found that extra-adrenal metastatic disease did not significantly alter either event free survival, or OS in a single-institution cohort.¹² However, in that study there was no differentiation between subjects with prior treated extra-adrenal metastatic disease and those with stable extra-adrenal disease at the time of adrenalectomy. Their findings are not incompatible with this study, and may reflect either excellent clinical judgement in selection of adrenalectomy patients with extra-adrenal metastatic disease, or effective systemic therapies, or a combination of the 2.

Our data shows that poorer survival outcomes are associated with residual cancer after adrenalectomy, either due to the presence of extra-adrenal oligometastatic disease or positive margins at the time adrenalectomy. Based on this finding, we suggest that extra-adrenal metastases at the time of adrenalectomy or adrenal disease invasive of adjacent tissues should be considered relative contraindications to therapeutic adrenal metastasectomy. This does not preclude adrenalectomy for palliation of symptoms or for other clinical considerations. Prior, treated extra-adrenal metastases should not be considered a contraindication to adrenalectomy.

Our findings recapitulate those of prior studies demonstrating no difference in oncologic outcomes or margin status between laparoscopic and open surgery.^{10,25,26} These data should be interpreted with caution, however, as outcomes were assessed retrospectively and based upon a priori clinical decision-making. It is worth noting that the tumor size was significantly larger in the subgroup undergoing open surgery and smallest in those undergoing laparoscopic resection, suggesting that tumor size factored into the choice of operative approach and; therefore, selection bias. Interestingly, tumor size was not independently associated with OS. Although tumor size was statistically significantly associated with disease free survival, the hazard ratio was low and therefore the clinical impact of tumor size on survival outcomes would be expected to be minimal. However, tumor size clearly influences operative approach. In our cohort, a minority of laparoscopic resections were performed from a posterior retroperitoneoscopic approach, and these procedures were institution-specific. Therefore, our data are insufficient to determine the potential benefits of laparoscopic transabdominal and retroperitoneoscopic approaches for adrenal metastasectomy. Within our study cohort, there was no statistically significant difference in margin status by operative approach ($P = 0.330$). However, the highest percentage of R2 margins (8%) was observed in the retroperitoneoscopic group, although this group had the smallest overall tumor size. In this retrospective series it is not possible to determine causality; however, it is conceivable that technical aspects related to restricted working space in the retroperitoneum may influence final surgical margins. The decision regarding operative approach should be determined based upon a global assessment of the patient's clinical status, anatomic and tumor considerations, and surgeon expertise, with the goal of a complete and margin-negative resection.

Percutaneous ablation techniques are a potential alternative to surgical resection of adrenal metastases. Percutaneous adrenal ablation procedures are performed for both benign and malignant disease; techniques include cryoablation, microwave ablation, and radiofrequency ablation.^{27,28} For large tumors (≥ 5 cm), ablation is less effective and therefore not routinely recommended.²⁹ It is worth noting that the median tumor size in our study was 4.5 cm, and; therefore, almost half of the patients who were treated with adrenalectomy in this cohort would not have been eligible for ablation.

Studies demonstrate variable effectiveness of adrenal ablation, with multiple ablations being required to achieve local control rates of approximately 80% at 1 year.^{27,28,30–35} To our knowledge, no randomized-control trials exist comparing oncologic outcomes of surgery versus ablation; this is an area which would benefit from more study. Nevertheless, the literature suggests that ablation is safe, and; therefore, could be considered for palliation and for patients who are not surgical candidates.

In our study, primary tumor type clearly impacted outcomes. Within the 5 most common primary tumor types, lung cancers were associated with the longest disease-free survival but the shortest OS. Patient demise followed closely after recurrence, consistent with the high mortality of Stage IV lung cancer. The rapid progression from recurrence to demise may also be reflective of the limited treatment options available for recurrent metastatic lung cancer. It is interesting to note that the median survival of subjects with lung cancer and resected adrenal metastases in this study was 47 months; in comparison, in an international cohort of 1059 patients with lung cancer and a single extra-thoracic site of metastatic disease, the median survival was 11.4 months.³⁶ Although selection bias may certainly contribute to the improved survival in the resected cohort, the survival advantage associated with adrenalectomy seems to be substantial and consistent with prior studies.^{7,9,22} Our data support the use of adrenalectomy as a treatment for metastatic lung cancer as it may improve survival.

In sarcoma, melanoma, and colorectal cancer, the converse pattern was observed, with relatively brief disease-free period after adrenalectomy, but longer survival. More than two-thirds of patients were alive 5 years after surgery. This may be reflective of highly effective systemic treatments. In the absence of a randomized clinical trial it would be challenging to prove a survival advantage to adrenalectomy in this clinical context. However as laparoscopic adrenalectomy is safe and well-tolerated,^{10,11} metastasectomy is a reasonable strategy for these patients.

This study is limited by its retrospective nature and sample size. Over the long period examined in this study, substantial changes in oncologic practice have led to heterogeneity of treatment and management. Additionally, as the rate of metastasectomy increased over the study period, the study cohort is necessarily skewed toward more recent surgical dates. Of the 269 patients in our study cohort, 101 underwent surgery in the last 3 years of the study. As a result, the duration of follow-up is necessarily limited, and survival outcomes are right-censored. It is also worth noting that the majority of sarcoma cases underwent operation at 2 centers, and therefore the results of this experience may not be broadly generalizable. Additionally, the number of colorectal cancer cases was small, and; therefore, these results must also be interpreted with caution. Although the overall cohort size is small, this is the largest study to date. These data capture the status of adrenal metastasectomy at representative tertiary institutions, and; therefore, have the potential for generalizability to oncologic practice.

Our findings suggest that oligometastatic disease at presentation is not a contraindication for adrenal metastasectomy, and even in the absence of cure, metastasectomy may provide a durable survival benefit. Based on these data, we suggest that adrenal metastasectomy be considered for patients with isolated adrenal metastases, regardless of prior treatment history. Adrenalectomy may confer a survival advantage in subjects with oligometastatic disease from primary lung cancer. Diverse primary tumor types may also benefit. Extra-adrenal oligometastatic disease at the time of adrenalectomy and direct invasion of adjacent structures by adrenal metastases should be considered relative contraindications for adrenal metastasectomy with curative intent. More research and prospective studies are needed to validate these indications.

ACKNOWLEDGMENT

The authors gratefully acknowledge the contributions of Jason A. Glenn, MD, to data collection.

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